



All Controllers for the General $\mathcal{H}_\infty$ Control Problem: LMI Existence Conditions and State Space Formulas*

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This paper solves the general $\mathcal{H}_\infty$ control problem by a purely algebraic approach. Existence conditions for an $\mathcal{H}_\infty$ controller are given in terms of linear matrix inequalities, and all $\mathcal{H}_\infty$ controllers are parametrized explicitly in state space.

Key Words—Linear systems; robust control; optimal control; matrix algebra; convex programming.

Abstract—This paper presents all controllers for the general $\mathcal{H}_\infty$ control problem (with no assumptions on the plant matrices). Necessary and sufficient conditions for the existence of an $\mathcal{H}_\infty$ controller of any order are given in terms of three Linear Matrix Inequalities (LMIs). Our existence conditions are equivalent to Scherer's results, but with a more elementary derivation. Furthermore, we provide the set of all $\mathcal{H}_\infty$ controllers explicitly parametrized in the state space using the positive definite solutions to the LMIs. Even under standard assumptions (full rank, etc.), our controller parametrization has an advantage over the Q-parametrization. The freedom Q (a real-rational stable transfer matrix with the $\mathcal{H}_\infty$ norm bounded above by a specified number) is replaced by a constant matrix L of fixed dimension with a norm bound, and the solutions (X, Y) to the LMIs. The inequality formulation converts the existence conditions to a convex feasibility problem, and also the free matrix L and the pair (X, Y) define a finite dimensional design space, as opposed to the infinite dimensional space associated with the Q-parametrization.

1. INTRODUCTION

IN THE LATE 1980s, $\mathcal{H}_\infty$ control theory reached a fairly mature state, complete with state space formulations (Glover and Doyle, 1988) and comprehensive comparisons with the widely known $\mathcal{H}_2$ (or LQG) control problem (Doyle *et al.*, 1989a). The existence conditions for an $\mathcal{H}_\infty$ controller were described by two Riccati equations and a spectral radius condition, and the set of all $\mathcal{H}_\infty$ controllers was parametrized using the (unique) stabilizing solutions to the

Riccati equations with free parameter Q, a real-rational stable transfer matrix with an $\mathcal{H}_\infty$ norm less than a specified number. Since there are many $\mathcal{H}_\infty$ controllers (beyond the central controller), the remaining freedom should be utilized to meet other specifications. However, a major deficiency of some of the existing $\mathcal{H}_\infty$ control theory as a practical tool for controller design is the fact that the free parameter Q is hard to choose [see the computational approaches in Boyd and Barratt (1991)]. The mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control problem has been solved directly (without optimizing over the free parameter Q) in Bernstein and Haddad (1989) and Doyle *et al.* (1989b) where necessary and sufficient conditions for optimality are obtained in terms of highly nonlinear coupled matrix equations, which are not easily solved. Khargonekar and Rotea (1991) overcame this difficulty via finite dimensional convex programming and obtained a feasible controller of order equal to the plant. Our paper is about $\mathcal{H}_\infty$ and not mixed problems. But, our result can be applied to solve the mixed problem directly, in exactly the same way as in Khargonekar and Rotea (1991), but we do not require any assumptions on the plant and the set of all feasible controllers (of any order) is parametrized explicitly.

Consider the n_p^{th} order linear time-invariant generalized plant

$$(\Sigma_p) \begin{cases} \dot{x} = Ax + B_1 w + B_2 u \\ z = C_1 x + D_{11} w + D_{12} u \\ y = C_2 x + D_{21} w + D_{22} u \end{cases} \quad (1)$$

where $x \in \mathbb{R}^{n_p}$ is the plant state, $w \in \mathbb{R}^{n_w}$ is any exogenous input, including plant disturbances,

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measurement noise, etc., $u \in \mathbb{R}^{n_u}$ is the control input, $z \in \mathbb{R}^{n_z}$ is the regulated output and $y \in \mathbb{R}^{n_y}$ is the measured output. Recall that most of the existing $\mathcal{H}_\infty$ control theory imposes the following “standard” assumptions: $D_{12}^T D_{12} > 0$ and $D_{21} D_{21}^T > 0$. There are a few exceptions. Petersen (1987) and Khargonekar *et al.* (1988) solved the singular state feedback case where $D_{12} = 0$ and $D_{21} = 0$ via the “algebraic Riccati equation approach”. Zhou and Khargonekar (1988) considered the state feedback case ($C_2 = I$, $D_{21} = 0$ and $D_{22} = 0$) without any assumption on the rank of D_{12} . Their approach for dealing with the singularity of $D_{12}^T D_{12}$ is basically to split the (constant) state feedback gain into two components; one belonging to the nullspace of D_{12} and the other orthogonal to it. As the existence condition for a controller, this approach yields an algebraic Riccati equation with a positive scalar parameter $\epsilon > 0$. To eliminate the extra work of searching for a suitable value of the parameter $\epsilon > 0$, Stoorvogel and Trentelman (1990) and Stoorvogel (1991) dealt with the singular $\mathcal{H}_\infty$ control problem more directly for the state feedback case and extended the result to include the dynamic output feedback case. In Stoorvogel and Trentelman (1990), and Stoorvogel (1991), assuming that the plant has no invariant zeros on the $j\omega$ -axis, the conditions for the existence of a controller were given in terms of Quadratic Matrix Inequalities (QMIs) and a set of rank conditions on the positive semidefinite solutions to the QMIs. It was shown that such solutions could be obtained as solutions to certain reduced-order Riccati equations. However, the author only gives a “conceptual algorithm” for constructing a controller of order equal to the generalized plant.

Further generalization is available from Scherer (1992a, b) for the state feedback case and for the dynamic output feedback case. The rank conditions and Stoorvogel’s assumption “no invariant zeros on the $j\omega$ -axis” are removed in Scherer’s work. The only restriction on the plant in Scherer (1992b) is that $D_{11} = 0$ and $D_{22} = 0$. In the specialized case $D_{11} = 0$, one of our results (the existence conditions in Theorem 4.2) is shown to be equivalent to the result of Scherer (1992b, Theorem 1). Scherer’s main focus was the existence of a controller (of any order) and the computation of the optimal $\mathcal{H}_\infty$ norm bound. For this purpose, he provided an algebraically verifiable method to check the existence of a solution to a Riccati inequality including a controller parameter, and gave only one (strictly proper) full order controller. Our focus is an explicit formula for parametrizing all

$\mathcal{H}_\infty$ controllers. Our result (Theorem 4.2) removes the dependence of Scherer’s Riccati inequality on the controller parameters, and shows the convexity of the set associated with the Riccati inequality solutions. In our formulation, the convex set is used to capture all $\mathcal{H}_\infty$ controllers, and is shown to be useful to incorporate additional performance specifications. Moreover, our new results provide the existence conditions for a fixed order $\mathcal{H}_\infty$ controller (e.g. static output feedback).

This paper considers a general $\mathcal{H}_\infty$ control problem without any assumptions on the plant matrices (A , B_1 , B_2 , C_1 , C_2 , D_{11} , D_{12} , D_{21}). We also make no assumption on D_{22} except for the well-posedness of the feedback connection. Consider the n_c^{th} order linear time-invariant dynamic ($n_c > 0$) and static ($n_c = 0$) controllers

$$(\Sigma_{cd}) \begin{cases} \dot{x}_c = A_c x_c + B_c y \\ u = C_c x_c + D_c y \end{cases}, \quad (\Sigma_{cs}) \{u = Gy, \quad (2)$$

where $x_c \in \mathbb{R}^{n_c}$ is the controller state. We shall denote the whole class of controllers ($n_c \geq 0$) by (Σ_c) . Suppose one defines a fictitious measured output $\hat{y} := C_2 x + D_{21} w$ of (1) (which has no term in u), and designs a controller (Σ_{cd}) assuming $\hat{y}$ is available for feedback, and then replaces $\hat{y}$ by $y - D_{22}u$. Then from (2),

$$u = (I + D_c D_{22})^{-1} (C_c x_c + D_c y) =: \tilde{C}_c x_c + \tilde{D}_c y \quad (3)$$

which is of the same form as (2). Of course, well-posedness assumes that $(I + D_c D_{22})^{-1}$ exists. Hence, under a well-posedness assumption, we need only a theory for treating (1) with $D_{22} = 0$. We consider the design of a stabilizing controller (Σ_c) which yields the closed-loop transfer matrix with $\mathcal{H}_\infty$ norm bounded above by a specified number. Such a controller is said to be an $\mathcal{H}_\infty$ controller. Necessary and sufficient conditions for the existence of an $\mathcal{H}_\infty$ controller of some (unspecified) order are given in terms of three Linear Matrix Inequalities (LMIs). Positive definite solutions to the LMIs form a convex set. The controller order can be fixed by imposing an additional rank condition (at the expense of convexity) on the solutions to the LMIs. Moreover, the set of all $\mathcal{H}_\infty$ controllers is characterized explicitly in the state space. Of course, the results can be specialized to the standard case. In this event, there is a strong similarity to the Q-parametrization (Doyle *et al.*, 1989a; Glover and Doyle, 1988) except that our existence conditions involve two Riccati inequalities instead of two Riccati equations, and the set of $\mathcal{H}_\infty$ controllers associated with a given solution pair to the Riccati inequalities is parametrized by a constant matrix of fixed dimension with a norm bound (this is a finite

dimensional parametrization) instead of the freedom in the Q-parametrization by any real-rational stable transfer matrix $Q(s)$ with an $\mathcal{H}_\infty$ norm bound (hence this is an infinite dimensional parametrization). Another difference is that positive definite solutions to Riccati inequalities are not unique and this nonuniqueness also provides freedom in the parametrization of all $\mathcal{H}_\infty$ controllers.

Our approach is purely algebraic [borrowed from the covariance control literature (Hotz and Skelton, 1987; Skelton and Iwasaki, 1993)]. Suppose an analysis results is given which states that a closed-loop system satisfies desirable specifications if and only if a matrix equation $M(P, G) = 0$ or a matrix inequality $M(P, G) < 0$ holds for some symmetric positive definite matrix P where G is a given controller parameter. The "covariance control approach" is to treat the underlying equation $M(P, G) = 0$ as an algebraic problem to be solved for the controller parameter G . In this case, solvability conditions are given in terms of P ; given P , there exists a controller G such that $M(P, G) = 0$ if and only if P belongs to a certain set $\mathcal{L}$. The set $\mathcal{L}$ is characterized without involving the controller parameter, and once we find a matrix $P \in \mathcal{L}$, controllers which achieve desired specifications can be obtained by an explicit (point-to-set) mapping of P . This separation of the controller parameter from $\mathcal{L}$ is the important contribution of the covariance control approach.

More recently, this algebraic approach has also been used in the $\mathcal{H}_\infty$ control literature. Sampei *et al.* (1990) used the strict bounded real lemma as the underlying inequality $M(P, G) < 0$, and, without any assumptions on the plant matrices except $D_{22} = 0$, reduced the problems of characterizing $\mathcal{L}$ to two state feedback problems. A class of full order controllers is given explicitly in terms of plant data and $P \in \mathcal{L}$. Gahinet (1992) considered the standard case and obtained a parametrization of fixed order $\mathcal{H}_\infty$ controllers where $\mathcal{L}$ is characterized by LMIs and a rank condition. However, after $P \in \mathcal{L}$ is obtained, additional problems involving inequalities have to be solved to obtain a controller since the parametrization is implicit. Skelton and Iwasaki (1993) solved a singular case and characterized $\mathcal{L}$ by three matrix equations similar to the "assignability conditions" in covariance control (Skelton and Iwasaki, 1993). An explicit formula for all fixed order $\mathcal{H}_\infty$ controllers is obtained. The present paper further develops the $\mathcal{H}_\infty$ control theory based on the purely algebraic approach introduced in covariance control. We will show that the general $\mathcal{H}_\infty$ control problem can be reduced to the problem of solving an LMI

for controller parameters. The set $\mathcal{L}$ is characterized by LMIs and a rank condition and all fixed order $\mathcal{H}_\infty$ controllers are parametrized explicitly using $P \in \mathcal{L}$. It will be shown that our existence conditions reduce to Gahinet's existence conditions when it is specialized appropriately.

The main advantage of our LMI formulation lies in the computational aspects; see Doyle *et al.* (1991) and Willems (1971) for related discussions. If the controller order is not fixed *a priori*, then the rank condition in the characterization of $\mathcal{L}$ can be removed and hence the computational problem becomes convex as has been shown in Gahinet (1992) for the standard case. Convexity of the set of all positive definite solutions to the LMIs implies the existence of powerful numerical algorithms to solve certain optimization or feasibility problems over the set (Beck, 1991; Geromel *et al.*, 1991). Finiteness of the dimension of the norm bounded design space means that a search for a particular value of the free parameters to give the desired closed-loop properties is much easier than a search in the infinite dimensional space of real-rational stable transfer matrices.

Another advantage of our LMI approach includes the fact that the controller order can be fixed by imposing a rank condition on the positive definite solutions to the LMI existence conditions. Of course, this additional rank condition would destroy convexity of the controller design problem, making the problem much harder; see Gahinet (1992), El Ghaoui and Gahinet (1993) and Iwasaki and Skelton (1993) for some computational approaches to this nonconvex problem. Nevertheless, our result parametrizes all low order controllers; if the controller order is not specified *a priori*, then the computational problem is convex and the resulting controller will be of order equal to or less than the plant.

The paper is organized as follows. Section 2 states the general $\mathcal{H}_\infty$ control problem and converts it to the problem of solving an LMI for controller parameters. Section 3 presents a linear algebra result which gives the necessary and sufficient conditions for the existence of a solution to a certain LMI and also parametrizes all solutions explicitly. Section 4 presents our main results. Conclusions are given in Section 5.

We will use the following notation. "If" means "if and only if". An $n \times m$ matrix with real elements is denoted by $A \in \mathbb{R}^{n \times m}$. A^T denotes the transpose of A . A^{-T} means $(A^T)^{-1}$. A^+ is the Moore–Penrose inverse of A . $\mathcal{N}(A)$ and $\mathcal{R}(A)$ denote the nullspace and the range space of A , respectively. $A^\perp$ denotes a matrix

with the following properties; $\mathcal{N}(A^\perp) = \mathcal{R}(A)$ and $A^\perp A^{\perp T} > 0$. Note that $A^\perp$ exists iff A has linearly dependent rows. Also note that, for a given A , $A^\perp$ is not unique but throughout the paper, any choice is acceptable. $\rho(A)$ is the spectral radius of A . The norm of a matrix $\|A\|$ is the largest singular value of A . For a symmetric matrix A , $\lambda_{\max}(A)$ denotes the largest eigenvalue. For a non-negative definite matrix A , $A^{1/2}$ denotes the unique non-negative definite square root of A . I_n is the $n \times n$ identity matrix. SVD stands for the singular value decomposition.

2. THE GENERAL $\mathcal{H}_\infty$ CONTROL PROBLEM

2.1. Problem formulation

Consider the linear time-invariant controller (Σ_c) and generalized plant (Σ_p) where $D_{22} = 0$. Combining the plant (Σ_p) and the static controller (Σ_{cs}) , the closed-loop system can be described by

$$(\Sigma_{cl}) \begin{cases} \dot{x} = A_{cl}x + B_{cl}w \\ z = C_{cl}x + D_{cl}w' \end{cases} \quad (4)$$

$$\begin{bmatrix} A_{cl} & B_{cl} \\ C_{cl} & D_{cl} \end{bmatrix} = \begin{bmatrix} A & B_1 \\ C_1 & D_{11} \end{bmatrix} + \begin{bmatrix} B_2 \\ D_{12} \end{bmatrix} G [C_2 \quad D_{21}]. \quad (5)$$

For the dynamic controller (Σ_{cd}) , the closed-loop system has exactly the same structure as for the static case, but the closed-loop state becomes $\hat{x}^T := [x^T \ x_c^T]$ and the plant matrices and the controller G in (5) are replaced with the following matrices with $\hat{\cdot}$ (hat):

$$\begin{bmatrix} \hat{A} & \hat{B}_1 & \hat{B}_2 \\ \hat{C}_1 & \hat{D}_{11} & \hat{D}_{12} \\ \hat{C}_2 & \hat{D}_{21} & \hat{G}^T \end{bmatrix} := \left[\begin{array}{cc|cc|c} A & 0 & B_1 & B_2 & 0 \\ 0 & 0 & 0 & 0 & I_{n_c} \\ \hline C_1 & 0 & D_{11} & D_{12} & 0 \\ C_2 & 0 & D_{21} & D_c^T & B_c^T \\ \hline 0 & I_{n_c} & 0 & C_c^T & A_c^T \end{array} \right]. \quad (6)$$

Hence, the fixed order dynamic controller design problem is a special case of the static controller design problem with a certain structure for the plant matrices as in (6).

We denote the closed-loop transfer matrix from w to z by T_{zw} . To state the $\mathcal{H}_\infty$ control problem, we need the following definition.

Definition 1. Given a scalar $\gamma > 0$. The controller (Σ_c) is said to be an " $\mathcal{H}_\infty$ controller" if the following two conditions hold:

- (1) A_{cl} is asymptotically stable.
- (2) $\|T_{zw}\|_\infty < \gamma$.

Note that the internal stability is included as a property of $\mathcal{H}_\infty$ controllers. Without loss of generality, we shall set the $\mathcal{H}_\infty$ norm bound γ to 1 to simplify the presentation for the rest of our discussion. Our objectives are to:

find the necessary and sufficient conditions for the existence of an $\mathcal{H}_\infty$ controller. If one exists, obtain an explicit formula for all such controllers.

2.2. LMI approach

In this subsection, we will convert the $\mathcal{H}_\infty$ control problem to a problem of solving an LMI for the controller parameter G or $\hat{G}$. The following lemma is essential in our formulation.

Lemma 1. Suppose a generalized plant (Σ_p) and a controller (Σ_c) are given. Define

$$Q := A_{cl}P + PA_{cl}^T + (PC_{cl}^T + B_{cl}D_{cl}^T)R^{-1} \times (PC_{cl}^T + B_{cl}D_{cl}^T)^T + B_{cl}B_{cl}^T, \quad (7)$$

$$R := I - D_{cl}D_{cl}^T. \quad (8)$$

The following statements are equivalent.

- (i) (Σ_c) is an $\mathcal{H}_\infty$ controller.
- (ii) $R > 0$ and there exists a $P > 0$ satisfying $Q < 0$.

Proof. This is a standard fact. See Zhou and Khargonekar (1988). $\square$

Recall that each closed-loop matrix (A_{cl} , B_{cl} , etc.) is a linear function of the controller parameter G for the static controller (Σ_{cs}) . Hence, the conditions $Q < 0$ and $R > 0$ are nonlinear matrix inequalities in G . Our approach for solving the $\mathcal{H}_\infty$ control problem is to find the necessary and sufficient conditions for the existence of a matrix G which satisfies the inequalities $Q < 0$ and $R > 0$ simultaneously, and gives an explicit parametrization of all such G when one exists. With such an approach, in general, existence conditions are obtained as inequalities with unknown matrix P . Thus the existence conditions for an $\mathcal{H}_\infty$ controller can be equivalently written in terms of the existence of a positive definite matrix P satisfying certain matrix inequalities. The matrix P will also be used to parametrize the set of all $\mathcal{H}_\infty$ controllers. For the dynamic controller (Σ_{cd}) , the same approach as above applies (indeed, the dynamic case result can be obtained by specializing the static case). In this case, the dimension of the matrix P fixes the order to the controller. However, our result shows that, with this algebraic approach, it is possible not to fix the controller order *a priori*.

The following lemma converts the two nonlinear matrix inequalities ($Q < 0$ and $R > 0$) into just one linear matrix inequality.

Lemma 2. Consider a generalized plant (Σ_p) and a static controller (Σ_{cs}). The following statements are equivalent.

- (i) The controller (Σ_{cs}) is an $\mathcal{H}_\infty$ controller.
- (ii) There exists a matrix $P > 0$ satisfying

$$BGC + (BGC)^T + \Omega < 0, \quad (9)$$

where

$$\begin{bmatrix} B & \Omega \\ * & C \end{bmatrix}' = \begin{bmatrix} B_2 & AP + PA^T & PC_1^T & B_1 \\ D_{12} & C_1P & -I_{n_z} & D_{11} \\ 0 & B_1^T & D_{11}^T & -I_{n_w} \\ * & C_2P & 0 & D_{21} \end{bmatrix}. \quad (10)$$

Proof. Recall that, for any matrices Φ_{11} , Φ_{12} and Φ_{22} where Φ_{11} and Φ_{22} are symmetric, the following are equivalent.

- (i) $\begin{bmatrix} \Phi_{11} & \Phi_{12} \\ \Phi_{12}^T & \Phi_{22} \end{bmatrix} > 0$
- (ii) $\Phi_{11} > 0$, $\Phi_{22} > \Phi_{12}^T \Phi_{11}^{-1} \Phi_{12}$
- (iii) $\Phi_{22} > 0$, $\Phi_{11} > \Phi_{12} \Phi_{22}^{-1} \Phi_{12}^T$.

Using this fact, Q and R defined in (7) and (8) satisfy $Q < 0$ and $R > 0$ iff

$$\begin{bmatrix} A_{cl}X_{cl} + X_{cl}A_{cl}^T + B_{cl}B_{cl}^T & X_{cl}C_{cl}^T + B_{cl}D_{cl}^T \\ C_{cl}X_{cl} + D_{cl}B_{cl}^T & D_{cl}D_{cl}^T - I_{n_z} \end{bmatrix} < 0. \quad (12)$$

After completing the square with respect to $[B_{cl}^T D_{cl}^T]$, we can use the same trick again to obtain an equivalent expression for (14) as follows:

$$\begin{bmatrix} A_{cl}X_{cl} + X_{cl}A_{cl}^T & X_{cl}C_{cl}^T & B_{cl} \\ C_{cl}X_{cl} & -I_{n_z} & D_{cl} \\ B_{cl}^T & D_{cl}^T & -I_{n_w} \end{bmatrix} < 0. \quad (13)$$

It is straightforward to show the equivalence of (9) and (13). $\square$

Lemma 2 reduced the general $\mathcal{H}_\infty$ control problem to a linear algebra problem of finding a matrix G satisfying (9). Needless to say, the dynamic controller case follows in a similar manner. The Schur complement formula (11) will be used frequently in the subsequent sections.

3. MATHEMATICAL PRELIMINARIES

This section presents a solution to the following LMI with unknown matrix G ,

$$BGC + (BGC)^T + Q < 0, \quad (14)$$

where Q is a symmetric matrix. The notation for all the matrices (B , Q , etc.) in this section is locally defined, that is, they have nothing to do with the matrices in the other sections (or lemmas within this section) even if the same symbols are used [for example, Q in (14) is different from Q in (7)]. As has been shown in the previous section, the LMI (14) represents the typical linear algebra problem which arises in the general $\mathcal{H}_\infty$ control problem. Our objective here is as follows. First, obtain necessary and sufficient conditions for the existence of a matrix G satisfying the LMI. Then provide an explicit formula for the set of all such G when one exists.

Before stating the result, we present a preliminary lemma and define two set of matrices, which will turn out to be the solution sets to the LMI (14).

Lemma 3. Let matrices $B \in \mathbb{R}^{n \times m}$ and $Q \in \mathbb{R}^{n \times n}$ be given. Suppose $\text{rank}(B) < n$ and $Q = Q^T$. Let (B_R, B_L) be any full rank factor of B , that is, $B = B_L B_R$, and define $D := (B_R B_R^T)^{-1/2} B_L^T$. Then

$$\mu B B^T - Q > 0 \quad (15)$$

holds for some $\mu \in \mathbb{R}$ iff

$$P := B^\perp Q B^{\perp T} < 0 \quad (16)$$

holds, in which case, all such μ are given by

$$\mu > \mu_{\min} := \lambda_{\max}[D(Q - Q B^{\perp T} P^{-1} B^\perp Q) D^T]. \quad (17)$$

Proof. Let T be a square nonsingular matrix defined by

$$T := \begin{bmatrix} D \\ B^\perp \end{bmatrix}. \quad (18)$$

By congruence transformation with T , (15) is equivalent to

$$\begin{bmatrix} \mu I - D Q D^T & -D Q B^{\perp T} \\ -B^\perp Q D^T & -B^\perp Q B^{\perp T} \end{bmatrix} > 0 \quad (19)$$

or equivalently,

$$P := B^\perp Q B^{\perp T} < 0 \quad (20)$$

$$\mu I - D Q D^T + D Q B^{\perp T} P^{-1} B^\perp Q D^T > 0 \quad (21)$$

which proves the necessity of (16). Now, to prove sufficiency, suppose $P < 0$. Obviously, there exists a μ satisfying (21) and all such μ are given by (17). $\square$

The existence condition (16) is known as Finsler's theorem [e.g. see references in Petersen and Hollot (1986)]. Lemma 3 provides not only the existence condition but also all acceptable values of μ . Note that $\mu_{\min} \leq 0$ iff

$Q \leq 0$ since $\mu_{\min} \leq 0$ is equivalent to

$$D(Q - QB^{\perp T}P^{-1}B^{\perp}Q)D^T \leq 0$$

which is equivalent to $TQT^T \leq 0$ since $P < 0$.

Definition 2. Let matrices $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{k \times n}$ and $Q = Q^T \in \mathbb{R}^{n \times n}$ be given. Let r_b and r_c be the ranks of B and C , respectively, and (B_L, B_R) and (C_L, C_R) be any full rank factors of B and C , that is, $B = B_L B_R$, $C = C_L C_R$. Then we define the following two sets:

$$\begin{aligned} \mathcal{G}_{\text{gen}}(B, C, Q) := & \left\{ G \in \mathbb{R}^{m \times k} : \exists (Z, L, R) \right. \\ & \left. \in \mathbb{R}^{m \times k} \times \mathbb{R}^{r_b \times r_c} \times \mathbb{R}^{r_b \times r_b} \right. \\ & \text{such that} \\ & G = B_R^+ K C_L^+ + Z - B_R^+ B_R Z C_L C_L^+ \\ & K := -R^{-1} B_L^T \Phi C_R^T (C_R \Phi C_R^T)^{-1} \\ & \quad + R^{-1} S^{1/2} L (C_R \Phi C_R^T)^{-1/2} \\ & \Phi := (B_L R^{-1} B_L^T - Q)^{-1} > 0, \\ & R > 0, \quad \|L\| < 1, \\ & S := R - B_L^T [\Phi - \Phi C_R^T C_R \Phi C_R^T]^{-1} B_L \left. \right\}, \quad (22) \end{aligned}$$

$$\begin{aligned} \mathcal{G}_{\text{std}}(B, C, Q) := & \left\{ G \in \mathbb{R}^{m \times k} : \exists L \in \mathbb{R}^{m \times k} \right. \\ & \text{such that} \\ & G = G_1 + G_2 L G_3, \quad \|L\| < 1, \\ & G_1 := (C_1^T - Q_{12} Q_{22}^{-1} C_2^T) (C_2 Q_{22}^{-1} C_2^T)^{-1}, \\ & G_2 := (Q_{12} Q_{22}^{-1} Q_{12}^T - Q_{11} - G_1 G_3^{-2} G_1^T)^{1/2}, \\ & G_3 := (-C_2 Q_{22}^{-1} C_2^T)^{-1/2}, \\ & \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{12}^T & Q_{22} \end{bmatrix} := \begin{bmatrix} B^+ \\ B^{\perp} \end{bmatrix} Q \begin{bmatrix} B^{+T} & B^{\perp T} \end{bmatrix}, \\ & [C_1 \quad C_2] := C \begin{bmatrix} B^{+T} & B^{\perp T} \end{bmatrix} \left. \right\}. \quad (23) \end{aligned}$$

Note that the above two sets may not be well-defined [e.g. the matrix $(-C_2 Q_{22}^{-1} C_2^T)$ in (23) may not be positive definite and G_3 is undefined] unless the triple (B, C, Q) satisfies certain conditions. This point will be made clear in the following theorem, which is the main result of this section.

Theorem 1. Let matrices $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{k \times n}$ and $Q = Q^T \in \mathbb{R}^{n \times n}$ be given. Suppose $\text{rank}(B) < n$ and $\text{rank}(C) < n$. Consider a set of matrices

$$\mathcal{G} := \{G \in \mathbb{R}^{m \times k} : BCG + (BCG)^T + Q < 0\}. \quad (24)$$

The following statements are equivalent.

- (i) $\mathcal{G} \neq \emptyset$.
- (ii) $B^{\perp} Q B^{\perp T} < 0$ and $C^{T\perp} Q C^{T\perp T} < 0$.

Suppose statement (ii) holds. Then the set $\mathcal{G}_{\text{gen}}(B, C, Q)$ is well-defined, and

$$\mathcal{G} \equiv \mathcal{G}_{\text{gen}}(B, C, Q).$$

If we further assume

$$B^T B > 0, \quad C B^{\perp T} B^{\perp} C^T > 0,$$

then the set $\mathcal{G}_{\text{std}}(B, C, Q)$ is well-defined, and

$$\mathcal{G} \equiv \mathcal{G}_{\text{std}}(B, C, Q).$$

Proof. Suppose $\mathcal{G} \neq \emptyset$ and let a matrix $G \in \mathcal{G}$ be given. Then $K := B_R G C_L$ satisfies

$$B_L K C_R + (B_L K C_R)^T + Q < 0.$$

Let a matrix $R > 0$ be such that

$$B_L K C_R + (B_L K C_R)^T + Q + C_R^T R K C_R < 0$$

or equivalently,

$$\begin{aligned} (K C_R + R^{-1} B_L^T)^T R (K C_R + R^{-1} B_L^T) \\ < B_L R^{-1} B_L^T - Q =: \Phi^{-1}. \quad (25) \end{aligned}$$

Note that such a matrix $R > 0$ always exists and a choice of such R is $R = \epsilon I$ for sufficiently small $\epsilon > 0$. Since the left-hand side of (25) is non-negative definite, we have $\Phi > 0$. Hence,

$$-B^{\perp} \Phi^{-1} B^{\perp T} = B^{\perp} Q B^{\perp T} < 0.$$

Now, by the Schur complement formula, (25) is equivalent to

$$R > (R K C_R + B_L^T) \Phi (R K C_R + B_L^T)^T.$$

After expanding the right-hand side, and completing the square with respect to RK , we have

$$\hat{L} (C_R \Phi C_R^T) \hat{L}^T < S, \quad (26)$$

where

$$\hat{L} := RK + B_L^T \Phi C_R^T (C_R \Phi C_R^T)^{-1}, \quad (27)$$

$$S := R - B_L^T [\Phi - \Phi C_R^T (C_R \Phi C_R^T)^{-1} C_R \Phi] B_L.$$

Again, since the left-hand side of (26) is nonnegative definite, we have $S > 0$. Note that

$$S > 0$$

$$\begin{aligned} & \Leftrightarrow \exists V > 0 : R > B_L^T [\Phi - \Phi C_R^T (C_R \Phi C_R^T \\ & \quad + V)^{-1} C_R \Phi] B_L \\ & \Leftrightarrow \exists V > 0 : R > B_L^T (\Phi^{-1} + C_R^T V^{-1} C_R)^{-1} B_L \\ & \Leftrightarrow \exists V > 0 : \Phi^{-1} - B_L R^{-1} B_L^T + C_R^T V^{-1} C_R > 0 \\ & \Leftrightarrow \exists V > 0 : C_R^T V^{-1} C_R - Q > 0. \quad (28) \end{aligned}$$

Hence, we have

$$C^{T\perp} (Q - C_R^T V^{-1} C_R) C^{T\perp T} = C^{T\perp} Q C^{T\perp T} < 0.$$

Note that (26) is equivalent to

$$\|L\| < 1, \quad L := S^{-1/2} \hat{L} (C_R \Phi C_R^T)^{1/2}.$$

Using the above definition of L , (27) can be solved for K as follows:

$$\begin{aligned} K := & -R^{-1} B_L^T \Phi C_R^T (C_R \Phi C_R^T)^{-1} \\ & + R^{-1} S^{1/2} L (C_R \Phi C_R^T)^{-1/2}. \end{aligned}$$

Finally, note that $K = B_R G C_L$ holds iff

$$G = B_R^+ K C_L^+ + Z - B_R^+ B_R Z C_L C_L^+$$

holds for some Z . Thus we have established the necessity of statements (ii).

To prove the converse, suppose (ii) holds, and consider the set $\mathcal{G}_{\text{gen}}(B, C, Q)$. From Finsler's theorem, $B^\perp Q B^{\perp T} < 0$ implies the existence of $R > 0$ such that $\Phi > 0$. Similarly, for such $R > 0$, $C^{T\perp} Q C^{T\perp T} < 0$ implies (28), and hence $S > 0$. Thus, (ii) implies that $\mathcal{G}_{\text{gen}}(B, C, Q)$ is well-defined. But clearly, $\mathcal{G} = \mathcal{G}_{\text{gen}}(B, C, Q)$ by construction.

Finally, we prove the last statement. Let a matrix $G \in \mathcal{G}$ be given. Then, by a congruent transformation, we have

$$\begin{bmatrix} B^+ \\ B^\perp \end{bmatrix} (BGC + (BGC)^T + Q) \begin{bmatrix} B^{+T} & B^{\perp T} \end{bmatrix} < 0,$$

or equivalently, using the definitions in (23),

$$\begin{bmatrix} Q_{11} + GC_1 + C_1^T G^T & Q_{12} + GC_2 \\ Q_{12}^T + C_2^T G^T & Q_{22} \end{bmatrix} < 0.$$

Since (ii) holds, we have $Q_{22} < 0$ and hence the above inequality is equivalent to

$$\begin{aligned} & Q_{11} + GC_1 + C_1^T G^T \\ & - (Q_{12} + GC_2) Q_{22}^{-1} (Q_{12} + GC_2)^T < 0. \end{aligned}$$

Now, by supposition,

$$C_2 C_2^T = C B^{\perp T} B^\perp C^T > 0,$$

and hence $C_2 Q_{22}^{-1} C_2^T < 0$. Thus, after expanding, we can complete the square as follows:

$$(G - G_1) G_3^{-2} (G - G_1)^T < G_2^2,$$

where matrices G_1 , G_2 and G_3 are defined in (23). The rest of the proof is similar to the above proof for more general case. This completes the proof. $\square$

4. ALL $\mathcal{H}_\infty$ CONTROLLERS

Necessary and sufficient conditions for the existence of an $\mathcal{H}_\infty$ controller and explicit state space formulas are obtained by directly applying the linear algebra result (Theorem 1) developed in the previous section to the general $\mathcal{H}_\infty$ control problem. First, we present our main result for static output feedback controllers. Recall that the fixed order dynamic output feedback problem has exactly the same mathematical structure as the static output feedback problem. Hence, the following theorem also provides all dynamic $\mathcal{H}_\infty$ controllers of fixed order by replacing the plant matrices with the augmented matrices defined in (6). To state the

result, we shall define the following two sets:

$$\begin{aligned} \mathcal{L}_B := & \left\{ X \in \mathbb{R}^{n_p \times n_p} : X = X^T > 0, \begin{bmatrix} B_2 \\ D_{12} \end{bmatrix}^\perp \right. \\ & \times \begin{bmatrix} AX + XA^T + B_1 B_1^T & XC_1^T + B_1 D_{11}^T \\ C_1 X + D_{11} B_1^T & D_{11} D_{11}^T - I \end{bmatrix} \\ & \left. \times \begin{bmatrix} B_2 \\ D_{12} \end{bmatrix}^{\perp T} < 0 \right\}. \quad (29) \end{aligned}$$

$$\begin{aligned} \mathcal{L}_C := & \left\{ Y \in \mathbb{R}^{n_p \times n_p} : Y = Y^T > 0, \begin{bmatrix} C_2^T \\ D_{21}^T \end{bmatrix}^\perp \right. \\ & \times \begin{bmatrix} YA + A^T Y + C_1^T C_1 & YB_1 + C_1^T D_{11} \\ B_1^T Y + D_{11}^T C_1 & D_{11}^T D_{11} - I \end{bmatrix} \\ & \left. \times \begin{bmatrix} C_2^T \\ D_{21}^T \end{bmatrix}^{\perp T} < 0 \right\}, \quad (30) \end{aligned}$$

if $(\cdot)^\perp$ exists in each set, otherwise, $\mathcal{L}_B$ and/or $\mathcal{L}_C$ are/is the set(s) of $n_p \times n_p$ real symmetric positive definite matrices. We also define a "dynamic version" of the above set, that is, let $\mathcal{L}_B$ and $\mathcal{L}_C$ be defined by replacing the plant matrices in the above definitions with the augmented matrices in (6). Note that, in this case, each set is a subset of $(n_p + n_c) \times (n_p + n_c)$ real symmetric matrices.

Theorem 2. The following statements are equivalent:

- (i) There exists a static output feedback $\mathcal{H}_\infty$ controller.
- (ii) $\mathcal{L}_S \neq \emptyset$ where

$$\mathcal{L}_S := \{ P \in \mathbb{R}^{n_p \times n_p} : P \in \mathcal{L}_B, P^{-1} \in \mathcal{L}_C \}.$$

Suppose $\mathcal{L}_S \neq \emptyset$ and let a matrix $G \in \mathbb{R}^{n_u \times n_y}$ be given. Then the following statements are equivalent:

- (i) G is a static output feedback $\mathcal{H}_\infty$ controller.
- (ii) $G \in \mathcal{G}_{\text{gen}}(B, C, \Omega)$ for some $P \in \mathcal{L}_S$,

where the matrix triple (B, C, Ω) is defined in (10).

Proof. Applying Theorem 1 to Lemma 2 directly yields the above result. The key steps are the following. We can choose

$$\begin{aligned} B^\perp &= \begin{bmatrix} \begin{bmatrix} B_2 \\ D_{12} \end{bmatrix}^\perp & 0 \\ 0 & I \end{bmatrix}, \\ C^{T\perp} &= \begin{bmatrix} \begin{bmatrix} C_2^T \\ D_{21}^T \end{bmatrix}^\perp & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} X^{-1} & 0 & 0 \\ 0 & 0 & I \\ 0 & I & 0 \end{bmatrix}, \end{aligned}$$

and use the Schur complement formula (i) $\Leftrightarrow$ (iii) in (11) for $B^\perp \Omega B^{\perp T} < 0$ and $C^{T\perp} \Omega C^{T\perp T} < 0$ to reduce the dimensions. $\square$

Note that the set $\mathcal{L}_S$, which describes the existence condition for a static $\mathcal{H}_\infty$ controller, is not convex. No method to find a matrix $P \in \mathcal{L}_S$ is available at this moment, and an algorithm is currently under development.

Next, we shall obtain a result for dynamic $\mathcal{H}_\infty$ controllers of arbitrary order based on the fixed order dynamic $\mathcal{H}_\infty$ controller results of Theorem 2.

Theorem 3. The following statements are equivalent:

- (i) There exists an $\mathcal{H}_\infty$ controller.
- (ii) $\mathcal{L}_D \neq \emptyset$ where

$$\mathcal{L}_D := \left\{ (X, Y) \in \mathbb{R}^{n_p \times n_p} \times \mathbb{R}^{n_p \times n_p} : \right.$$

$$\left. X \in \mathcal{L}_B, Y \in \mathcal{L}_C, \begin{bmatrix} X & I \\ I & Y \end{bmatrix} \geq 0 \right\}.$$

Suppose $\mathcal{L}_D \neq \emptyset$, and a matrix $\hat{G}$ be given. Then the following statements are equivalent:

- (i) $\hat{G}$ is an $\mathcal{H}_\infty$ controller.
- (ii) $\hat{G} \in \mathcal{G}_{\text{gen}}(\hat{B}, \hat{C}, \hat{\Omega})$ for some P where

$$P := \begin{bmatrix} X & P_{12} \\ P_{12}^T & P_{22} \end{bmatrix}, \quad P_{12}P_{22}^{-1}P_{12}^T := X - Y^{-1},$$

$$P_{22} > 0, \quad (X, Y) \in \mathcal{L}_D,$$

where the matrix triple $(\hat{B}, \hat{C}, \hat{\Omega})$ is defined in (11) by replacing the plant matrices with the augmented matrices defined in (6).

Proof. Recall that Theorem 2 can be specialized to the case for fixed order dynamic $\mathcal{H}_\infty$ controllers of order n_c if we replace the plant matrices with the augmented matrices defined in (6). Consider a set $\hat{\mathcal{L}}_S$, which is a version of $\mathcal{L}_S$ modified with the augmented matrices, and let a positive definite matrix $P \in \mathbb{R}^{(n_p+n_c) \times (n_p+n_c)}$ be given. Defining appropriately partitioned blocks of P by

$$\begin{bmatrix} X & P_{12} \\ P_{12}^T & P_{22} \end{bmatrix} := P, \quad Y := (X - P_{12}P_{22}^{-1}P_{12}^T)^{-1} > 0 \quad (31)$$

and noting that

$$\begin{bmatrix} \hat{B}_2 \\ \hat{D}_{12} \end{bmatrix}^\perp = \begin{bmatrix} [B_2]^\perp \\ [D_{12}]^\perp \end{bmatrix} \begin{bmatrix} I_{n_p} & 0 & 0 \\ 0 & 0 & I_{n_z} \\ 0 & I_{n_c} & 0 \end{bmatrix},$$

$$\begin{bmatrix} \hat{C}_2^T \\ \hat{D}_{21}^T \end{bmatrix}^\perp = \begin{bmatrix} [C_2^T]^\perp \\ [D_{21}^T]^\perp \end{bmatrix} \begin{bmatrix} I_{n_p} & 0 & 0 \\ 0 & 0 & U_{n_w} \\ 0 & I_{n_c} & 0 \end{bmatrix},$$

it is easy to see that $P \in \hat{\mathcal{L}}_S$ iff $X \in \mathcal{L}_B$ and $Y \in \mathcal{L}_C$. Conversely, for given positive definite matrices $X \in \mathcal{L}_B$ and $Y \in \mathcal{L}_C$, a positive definite

matrix P such that (31) holds can be constructed iff $X - Y^{-1} \geq 0$, in which case, such a P satisfies $P \in \hat{\mathcal{L}}_S$. Finally, noting that

$$X \geq Y^{-1} > 0 \Leftrightarrow \begin{bmatrix} X & I \\ I & Y \end{bmatrix} \geq 0,$$

we conclude the result. $\square$

We shall show the connection between our existence condition $\mathcal{L}_D \neq \emptyset$ and the previous work by Scherer (1992b). In the sequel, we assume $D_{11} = 0$ as has been done in Scherer (1992b). Suppose all the states are available for feedback, that is, $C_2 = I$ and $D_{21} = 0$. Then the inequality in the definition of $\mathcal{L}_C$ becomes redundant, and $\mathcal{L}_C$ is just the set of real symmetric positive definite matrices. Hence, for any $X \in \mathcal{L}_B$, we can always choose Y to be X^{-1} to satisfy the third condition (LMI) in $\mathcal{L}_D$, which implies from Theorem 2 a known fact (Khargonekar *et al.*, 1988) that a state feedback $\mathcal{H}_\infty$ controller can always (if one exists) be chosen to be a static gain. In this case, noting that $\mathcal{L}_D \neq \emptyset$ iff $\mathcal{L}_B \neq \emptyset$, we conclude the following:

$$\mathcal{L}_B = \{X \in \mathbb{R}^{n_p \times n_p} : X = X^T > 0, \exists F \text{ such that}$$

$$(A + B_2F)X + X(A + B_2F)^T + X(C_1 + D_{12}F)^T$$

$$\times (C_1 + D_{12}F)X + B_1B_1^T < 0\}.$$

Now we can see the equivalence of our existence condition $\mathcal{L}_D \neq \emptyset$ and Scherer's result (1992b, Theorem 1). Specifically, his set $\mathcal{P}_\mu$, which describes the existence condition, may be written as

$$\mathcal{P}_\mu = \{P \in \mathbb{R}^{n_p \times n_p} : P^{-1} \in \mathcal{L}_B\},$$

and the same argument applies to the "dual" set $\mathcal{L}_C$, and the last condition in $\mathcal{L}_C$ can be shown to be equivalent to Scherer's spectral radius condition $\rho(X^{-1}Y^{-1}) < 1$ except for the strict inequality. Scherer showed that the existence condition $\mathcal{P}_\mu \neq \emptyset$ can be checked by solving a reduced order Riccati equation, for the purpose of computing the optimal $\mathcal{H}_\infty$ norm bound. Our contribution is that the set of matrices X such that $X^{-1} \in \mathcal{P}_\mu$ is shown to be convex and characterized by $\mathcal{L}_B$, for the purpose of capturing all $\mathcal{H}_\infty$ controllers.

Theorem 3 gives an explicit state space parametrization of the set of all $\mathcal{H}_\infty$ controllers of arbitrary (finite) order using the positive definite matrix pair $(X, Y) \in \mathcal{L}_D$. Clearly, the set $\mathcal{L}_D$ is convex since every constraint is linear. Hence, the existence of an $\mathcal{H}_\infty$ controller may be checked via finite dimensional convex feasibility programming. However, Scherer's (1992a, b) reduced order algebraic Riccati equation formulation works better for this purpose since it

requires no iteration. Convexity of the set $\mathcal{L}_D$ is most useful when we try to optimize a certain performance measure, which can be expressed in terms of a convex function of $(X, Y) \in \mathcal{L}_D$, over the set of all $\mathcal{H}_\infty$ controllers. Mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control (Bernstein *et al.*, 1939; Khargonekar and Rotea, 1991) belongs to this class of problems. Specifically, if we define an error signal $e := C_0 x$, then the $\mathcal{H}_2$ performance bound (Bernstein *et al.*, 1989) with respect to the transfer matrix from w to e may be given by

$$J = \text{trace}(C_0 X C_0^T).$$

Thus we can pose the following mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control problem;

$$\inf \{J : (X, Y) \in \mathcal{L}_D\},$$

which can be solved via finite dimensional convex optimization program as in Khargonekar and Rotea (1991). This approach can be extended easily to other performances such as the $\mathcal{L}_2$ to $\mathcal{L}_\infty$ gain bound and robust $\mathcal{H}_2$ guaranteed cost.

Another effective utilization of the set $\mathcal{L}_D$ may be found in the computation of the optimal $\mathcal{H}_\infty$ norm. To this end, let the matrices $[C_1 \ D_{11} \ D_{12}]$ be replaced by $\gamma^{-1}[C_1 \ D_{11} \ D_{12}]$ in (29) and (30) to recover the γ -dependence, and define the corresponding sets $(X, \gamma^2) \in \mathcal{L}_{B_\gamma}$, $(Y, \gamma^2) \in \mathcal{L}_{C_\gamma}$. Then it is easy to see that the set $\mathcal{L}_{B_\gamma}$ is convex, and it can be shown using the Schur complement formula that the set $\mathcal{L}_{C_\gamma}$ is also convex. Hence, the set $(X, Y, \gamma^2) \in \mathcal{L}_D$, corresponding to $\mathcal{L}_D$ with γ explicitly expressed, is convex. Then the optimal $\mathcal{H}_\infty$ norm bound may be found by solving the following convex optimization problem:

$$\inf \{\gamma^2 : (X, Y, \gamma^2) \in \mathcal{L}_D\}.$$

Such a convex programming approach has been taken in Peres *et al.* (1993) for the state feedback case with the standard assumption. Our LMI approach generalizes the result of Peres *et al.* (1993) to the dynamic output feedback case with no assumptions on the plant.

The controller order n_c is not fixed *a priori* in the parametrization of all $\mathcal{H}_\infty$ controllers in Theorem 3. Recall that the order n_c is related to the dimension of the matrix P in (9); $P \in \mathbb{R}^{(n_p+n_c) \times (n_p+n_c)}$. Hence, in view of Theorem 3, the order n_c is determined by the dimension of $P_{22} \in \mathbb{R}^{n_c \times n_c}$ when finding matrices P_{12} and P_{22} such that

$$P_{12} P_{22}^{-1} P_{12}^T = X - Y^{-1}, \quad P_{22} > 0$$

for a given pair $(X, Y) \in \mathcal{L}_D$. Note that for any given $(X, Y) \in \mathcal{L}_D$, we have $X \geq Y^{-1}$ due to the third condition in the definition of $\mathcal{L}_D$. Hence,

the above factorization is always possible. Moreover, the lowest possible controller order for a given pair $(X, Y) \in \mathcal{L}_D$ is achieved when the factorization is such that P_{12} has full column rank, which is equal to $\text{rank}(X - Y^{-1})$. Thus, adding another condition

$$\text{rank}(X - Y^{-1}) = n_c$$

in the definition of $\mathcal{L}_D$, we may obtain the existence condition for a fixed order $\mathcal{H}_\infty$ controller, which is equivalent to, but different from, the dynamic version of the existence condition in Theorem 2. It is worth noting that for any $\mathcal{H}_\infty$ controller of order $n_c > n_p$, one can always construct a controller of order $n_c = n_p$ which achieves the same $\mathcal{H}_\infty$ norm bound. This known fact [see, for example, Doyle *et al.* (1989a)] can easily be verified by noting that the rank of $(X - Y^{-1})$ can never exceed n_p due to the dimension constraint.

Overall, we have the following controller design freedom: $(X, Y) \in \mathcal{L}_D$, the factorization to P_{12} and P_{22} , the matrix triple (Z, L, R) in (22). The freedom due to Z disappears if there are no redundant actuators ($B_2^T B_2 > 0$) and no redundant measurements ($C_2 C_2^T > 0$). It is clear from the proof of Theorem 1 that the positive definite matrix R can be restricted to have the structure of $R = \epsilon I$, where ϵ is a real number, without loss of generality. It can be shown by Lemma 3 that the admissible interval of the scalar ϵ is $0 < \epsilon < \epsilon_{\max}$, where $\epsilon_{\max}$ is a computable (without iteration) positive scalar or infinity. Hence, all the essential controller design freedoms are $(X, Y) \in \mathcal{L}_D$, $\|L\| < 1$ and $\epsilon > 0$. We have discussed intensively the use of the freedom due to $(X, Y) \in \mathcal{L}_D$. For a given $(X, Y) \in \mathcal{L}_D$, it is possible to characterize all corresponding (fixed order) $\mathcal{H}_\infty$ controllers by the LMI (9), where P has been determined from (X, Y) as in Theorem 3. Such a parametrization is useful when we want to optimize some performance expressed in terms of $\hat{G}$ since the LMI (9) defines a convex set of matrices $\hat{G}$. However, since important performance specifications such as $\mathcal{H}_2$ norm bound are taken into account by the choice of $(X, Y) \in \mathcal{L}_D$, it is nice to compute a controller corresponding to a given (X, Y) without iteration. This is made possible by the explicit characterization of $\mathcal{H}_\infty$ controllers given by $\mathcal{G}_{\text{gen}}(\hat{B}, \hat{C}, \hat{\Omega})$.

Finally, we shall show how our general results (Theorem 3) can be specialized with the following standard assumptions:

$$\begin{aligned} D_{11} &= 0, & D_{12}^T [C_1 \ D_{12}] &= [0 \ I], \\ D_{21} [B_1^T \ D_{21}^T] &= [0 \ I]. \end{aligned} \quad (32)$$

Theorem 4. Suppose the Assumption (32) holds. Then the following statements are equivalent.

- (i) There exists an $\mathcal{H}_\infty$ controller.
- (ii) $\mathcal{L}_{D\text{std}} \neq \emptyset$ where

$$\begin{aligned} \mathcal{L}_{D\text{std}} &:= \{(\bar{X}, \bar{Y}) \in \mathbb{R}^{n_p \times n_p} \times \mathbb{R}^{n_p \times n_p} : \\ &\quad \bar{X} = \bar{X}^T > 0, \bar{Y} = \bar{Y}^T > 0, \\ &\quad \bar{X}A + A^T\bar{X} + \bar{X}(B_1B_1^T - B_2B_2^T)\bar{X} + C_1^TC_1 < 0, \\ &\quad A\bar{Y} + \bar{Y}A^T + \bar{Y}(C_1^TC_1 - C_2^TC_2)\bar{Y} + B_1B_1^T < 0, \\ &\quad \rho(\bar{X}\bar{Y}) \leq 1\}. \end{aligned}$$

Suppose $\mathcal{L}_{D\text{std}} \neq \emptyset$, and a matrix $\hat{G}$ be given. Then the following statements are equivalent.

- (i) $\hat{G}$ is an $\mathcal{H}_\infty$ controller.
- (ii) $\hat{G} \in \mathcal{G}_{\text{std}}(\hat{B}, \hat{C}, \hat{\Omega})$ for some P where

$$P := \begin{bmatrix} \bar{X}^{-1} & P_{12} \\ P_{12}^T & P_{22} \end{bmatrix},$$

$$P_{12}P_{22}^{-1}P_{12}^T := \bar{X}^{-1} - \bar{Y}, \quad P_{22} > 0, \quad (\bar{X}, \bar{Y}) \in \mathcal{L}_{D\text{std}},$$

where the matrix $(\hat{B}, \hat{C}, \hat{\Omega})$ is defined in (10) by replacing the plant matrices with the augmented matrices defined in (6).

Proof. With the Assumption (32), we can choose

$$\begin{aligned} \begin{bmatrix} B_2 \\ D_{12} \end{bmatrix}^\perp &= \begin{bmatrix} I_{n_p} & -B_2D_{12}^+ \\ 0 & D_{12}^\perp \end{bmatrix}, \\ \begin{bmatrix} C_2^T \\ D_{21}^T \end{bmatrix}^\perp &= \begin{bmatrix} I_{n_p} & -(D_{21}^+C_2)^T \\ 0 & D_{21}^{T\perp} \end{bmatrix}, \end{aligned}$$

and substitute into (29) and (30). Then, using the Schur complement formula and defining $\bar{X} := X^{-1}$ and $\bar{Y} := Y^{-1}$, it is easy to obtain the Riccati inequalities in $\mathcal{L}_{D\text{std}}$. The third (LMI) condition in $\mathcal{L}_D$ is equivalent to $X \geq Y^{-1} > 0$, which is equivalent to $X > 0$, $Y > 0$ and $\rho(X^{-1}Y^{-1}) \leq 1$. This proves the existence condition.

To prove the controller formula, recall that we are solving the LMI (9) with augmented matrices $(\hat{B}, \hat{C}, \hat{\Omega})$, using Theorem 1. Let $P \in \mathbb{R}^{(n_p+n_c) \times (n_p+n_c)}$ be partitioned as

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^T & P_{22} \end{bmatrix}, \quad (33)$$

and using a choice of $\hat{B}^\perp$:

$$\hat{B}^\perp := \begin{bmatrix} I_{n_p} & 0 & -B_2D_{12}^+ & 0 \\ 0 & 0 & D_{12}^\perp & 0 \\ 0 & 0 & 0 & I_{n_w} \end{bmatrix},$$

we have

$$\begin{aligned} \hat{B}^T\hat{B} &= \begin{bmatrix} B_2^TB_2 + I_{n_u} & 0 \\ 0 & I_{n_c} \end{bmatrix}, \\ \hat{C}\hat{B}^{\perp T} &= \begin{bmatrix} C_2P_{11} & 0 & D_{21} \\ P_{12}^T & 0 & 0 \end{bmatrix}. \end{aligned}$$

Clearly, $\hat{B}^T\hat{B} > 0$, and $\hat{C}\hat{B}^{\perp T}$ has linearly independent rows, that is, $\hat{C}\hat{B}^{\perp T}\hat{B}^\perp\hat{C}^T > 0$ iff $P_{12}^TP_{12} > 0$ since $D_{21}D_{21}^T > 0$. But we can assume $P_{12}^TP_{12} > 0$ without loss of generality due to the following reason: let any $\mathcal{H}_\infty$ controller $\hat{G}$ be given. Then there exists a matrix $P > 0$ satisfying (9) with the augmented matrices (6). Let P be partitioned as in (33) and suppose $P_{12}^TP_{12} \not> 0$. Define $\tilde{P}_{12}$ by a matrix obtained by replacing the zero singular value(s) of P_{12} with $\epsilon > 0$ in the SVD of P_{12} . Correspondingly, define $\tilde{P}$ by replacing the 12-block P_{12} of P with $\tilde{P}_{12}$. Then, for sufficiently small $\epsilon > 0$, we have $\tilde{P}_{12}^T\tilde{P}_{12} > 0$, $\tilde{P} > 0$ and $\tilde{P}$ satisfies (9). Thus, all $\mathcal{H}_\infty$ controllers can still be captured even if we restrict our attention to the class of matrices $P > 0$ with the property $P_{12}^TP_{12} > 0$. The rest of the proof directly follows from the latter half of Theorem 1.

The existence condition $\mathcal{L}_{D\text{std}} \neq \emptyset$ obtained by specializing Theorem 3 is equivalent to the central result of Gahinet (1992). Explicit formulas for all $\mathcal{H}_\infty$ controllers can also be obtained by specializing Theorem 3 to the standard case. However, we can eliminate the free parameter $R > 0$ from the controller formula if we directly consider the standard case. Notice that our parameterization of all $\mathcal{H}_\infty$ controllers has similarity to the well-known state space formula for the standard $\mathcal{H}_\infty$ control problem (Doyle *et al.*, 1989a) in the following sense. Recall that in the Q-parametrization, the free parameter is a real-rational transfer matrix with its $\mathcal{H}_\infty$ norm bounded above by a specified number. In our case, the freedom is a constant matrix of fixed dimension with a norm bound. It should be noted that our design freedom also includes $(\bar{X}, \bar{Y}) \in \mathcal{L}_{D\text{std}}$ as well as $\|L\| < 1$, whereas the Riccati equation solutions in the Q-parametrization are unique and all the freedom is captured by a single parameter Q , which defines an infinite dimensional convex freedom set. Although each of the freedoms $(\bar{X}^{-1}, \bar{Y}^{-1})$ and L is convex, the combined total design freedom set $(\bar{X}^{-1}, \bar{Y}^{-1}, L)$ is not convex but finite dimensional.

For the singular case where both D_{12} and D_{21} are zero matrices, we can also specialize the general result (Theorem 3) to obtain the ϵ -Riccati equation by Petersen (1987).

5. CONCLUSION

The solution to the general $\mathcal{H}_\infty$ control problem is presented. Existence conditions are given in terms of three linear matrix inequalities. The set of all $\mathcal{H}_\infty$ controllers is parametrized using positive definite solutions to the LMIs,

which form a convex set. For each pair of the positive definite matrices, there corresponds a set of $\mathcal{H}_\infty$ controllers. A state space formula parametrizing all such controllers is obtained, where essential free parameters are constant matrices of fixed dimensions with a norm bound and a real number ϵ within a known interval.

If we specialize our result to the standard case where D_{12} and D_{21} are full rank, the general LMI existence conditions reduce to Riccati inequalities which are identical to the familiar $\mathcal{H}_\infty$ Riccati equations with minor modifications (such as replacement of inequalities by equalities). The set of all controllers associated with given positive definite matrices satisfying the Riccati inequalities is parametrized by an arbitrary matrix of fixed dimension with a norm bound.

The main advantages of our results over the existing $\mathcal{H}_\infty$ control theory are as follows. We do not require any assumptions on the generalized plant matrices (except for the well-posedness of the feedback connection). Our parametrization has computational advantages due to the free design parameters L and (X, Y) , each (but not all) of which forms a finite dimensional convex set (note that the Q -parametrization is convex as a whole but not finite dimensional). Furthermore, the controller order can be fixed if desired, at the expense of convexity.

Possible directions to which our results can be extended include the general $\mathcal{H}_\infty$ control problem for discrete-time systems and linear time-varying systems (Ravi *et al.*, 1991), robust control problems with quadratic stabilization [see Khargonekar *et al.* (1990) and references therein], mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control problems (Khargonekar and Rotea, 1991), multiple objective and/or the generalized $\mathcal{H}_2$ control problems (Rotea, 1993), and covariance (upper bound) assignment problems (Skelton and Iwasaki, 1993).

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